



Federated learning for prenatal detection of interrupted aortic arch using fetal ultrasound imaging

Jiancheng Han^{a,b,1}, Heqing Wang^{c,d,1}, Yifan Feng^{c,d,1}, Qi Yang^{a,b,1}, Jingtian Li^{c,d}, Haojie Zhang^{c,d}, Yihua He^{a,b}, Jiang Liu^e, Toru Nakamura^f, Yang Cao^g, Naidi Sun^{c,d,*}, Kun Qian^{c,d,*}, Bin Hu^{c,d,*}, Xinru Gao^h, Yan Xiaⁱ, Zongjie Weng^j, Björn W. Schuller^{k,l}, Yoshiharu Yamamoto^m

^a Echocardiography Medical center, Beijing Anzhen Hospital, Capital Medical University, Beijing, China

^b Beijing Key Laboratory of Maternal-Fetus Medicine in Fetal Heart Disease, Beijing, China

^c Key Laboratory of Brain Health Intelligent Evaluation and Intervention (Beijing Institute of Technology), Ministry of Education, Beijing, China

^d School of Medical Technology, Beijing Institute of Technology, Beijing, China

^e Global center for Science and Engineering, Faculty of Science and Engineering, Waseda University, Tokyo, Japan

^f Graduate School of Engineering Science, The University of Osaka, Osaka, Japan

^g Institute of Science Tokyo, Tokyo, Japan

^h Northwest Women's and Children's Hospital, Xi'an, China

ⁱ Peking University First Hospital Ningxia Women and Children's Hospital (Ningxia Hui Autonomous Region Maternal and Child Health Hospital), Yinchuan, China

^j Department of Medical Ultrasonics, Fujian Maternity and Child Health Hospital, College of Clinical Medicine for Obstetrics & Gynecology and Pediatrics, Fujian Medical University, Fuzhou, China

^k GLAM – the Group on Language, Audio, & Music, Imperial College London, London, UK

^l CHI – the Chair of Health Informatics, TUM University Hospital, Munich, Germany

^m Educational Physiology Laboratory, Graduate School of Education, The University of Tokyo, Tokyo, Japan

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ABSTRACT

This study presents the first application of federated learning (FL) for prenatal detection of Interrupted Aortic Arch (IAA) using fetal ultrasound images. To address the challenges of data scarcity, privacy constraints, and inter-institutional variability, we develop a federated learning IAA detection method and systematically evaluate three representative strategies (FedAvg, FedProx, and FedBABU) across five clinical centres. Results show that FL improves model performance over local training in recall and F1-score in data-scarce centres. Among FL algorithms, FedAvg and FedProx consistently outperform FedBABU in stability and generalisation. Among the three CNN architectures compared — ResNet-50, EfficientNet-B3, and DenseNet-121 — DenseNet-121 demonstrates superior overall performance, particularly in non-independent and identically distributed (Non-IID) scenarios. Our framework demonstrates the feasibility of collaborative AI for rare disease detection without data sharing, laying the foundation for scalable, real-world prenatal screening of congenital heart defects.

1. Introduction

Interrupted Aortic Arch (IAA) is a rare yet severe congenital heart defect characterised by a complete anatomical discontinuity between the ascending and descending aorta [1]. Despite accounting for less than 1% of congenital heart disease (CHD) cases, IAA remains a significant prenatal screening concern due to its clinical severity and the high birth rates in countries like China and India [2]. In the

absence of timely diagnosis and surgical intervention, IAA often results in profound circulatory failure shortly after birth, leading to high rates of neonatal mortality [1]. Nevertheless, owing to its low incidence, individual medical centres typically encounter only a limited number of cases, posing a major obstacle to developing accurate and generalisable diagnostic models through conventional data-driven methods [3–5]. Moreover, IAA encompasses distinct anatomical subtypes with heterogeneous clinical presentations, further complicating prenatal diagnosis.

* Corresponding authors.

E-mail addresses: sunnaidi@bit.edu.cn (N. Sun), qian@bit.edu.cn (K. Qian), bh@bit.edu.cn (B. Hu).

¹ Jiancheng Han, Heqing Wang, Yifan Feng, and Qi Yang contributed equally to this paper. They should be considered as co-first authors.

Clinically, IAA is not a uniform entity but comprises several anatomical subtypes with distinct clinical implications. According to the anatomical site of interruption, IAA is commonly classified into three subtypes [6]: Type A, in which the interruption occurs distal to the left subclavian artery; Type B, where the interruption is located between the left common carotid artery and the left subclavian artery; and Type C, characterised by an interruption proximal to the left common carotid artery. Among these, Type B is the most frequently encountered subtype in clinical practice and is often associated with additional cardiac anomalies.

As one of the most critical congenital heart diseases, IAA requires immediate neonatal intervention after birth. Without timely diagnosis and surgical treatment, affected neonates may rapidly develop severe congestive heart failure, with reported neonatal mortality rates reaching up to 90% [7]. Therefore, accurate prenatal identification of IAA is crucial for perinatal management, delivery planning, and postnatal surgical outcomes. Fetal echocardiography remains the primary modality for prenatal screening and diagnosis of IAA [7]. However, due to operator dependency, anatomical complexity, and substantial inter-institutional variability, the diagnostic performance for IAA remains highly heterogeneous across centres, with reported prenatal detection rates ranging from 0% to 100% [8].

In addition to its inherent clinical complexity, the low incidence of IAA further presents fundamental challenges for data-driven diagnostic modelling. Unlike common diseases, for which large datasets can often be assembled within a single institution, rare congenital anomalies such as IAA necessitate data aggregation from multiple clinical centres to train reliable and generalisable models. However, such aggregation is difficult to achieve in practice due to strict data protection protocols, institutional privacy policies, and ethical considerations surrounding the use of sensitive medical information, particularly in fetal and pediatric populations [9–11]. These factors lead to isolated data repositories across institutions, resulting in fragmented datasets that are insufficient for robust model development [12]. For rare conditions where each case is critical, this lack of data integration substantially hinders the potential of artificial intelligence to assist in early and accurate diagnosis [4,13].

To address these challenges, federated learning (FL) has emerged as a promising solution. FL is a decentralised machine learning paradigm that allows multiple institutions to collaboratively train AI models without exchanging raw patient data [14]. Instead, each institution retains its local dataset and computes model updates independently. Only encrypted model parameters—such as gradients or weights—are transmitted to a central server, which aggregates them to update a global model. This approach preserves patient privacy and complies with data protection laws while enabling cross-institutional collaboration [15,16]. For rare disease modelling, where pooling data is essential yet often prohibited, FL offers a powerful mechanism to overcome data scarcity while respecting ethical and legal constraints [17–19].

In the context of prenatal diagnostics, fetal ultrasound imaging is the primary modality used to screen for structural heart defects, including IAA [20]. Ultrasound is widely accessible, real-time, and non-invasive, making it ideal for fetal examinations. Standard cardiac views—such as the four-chamber and outflow tract views—are routinely acquired during obstetric scans [21]. However, the interpretation of these images is highly operator-dependent and technically challenging [22]. Image quality can be degraded by fetal motion, acoustic shadowing, and poor contrast, and detecting complex anomalies like IAA requires significant expertise [23]. Even experienced sonographers may miss subtle abnormalities, leading to missed or delayed diagnoses [24]. These factors underscore the critical need for robust AI systems that can assist clinicians by providing consistent and accurate interpretation of fetal cardiac ultrasound images.

Despite the promise of AI in prenatal imaging, existing models are often developed using single-centre datasets, which are inherently limited in size and diversity—especially for rare conditions [25]. Such

models tend to perform well on internal data but fail to generalise across institutions due to differences in imaging protocols, machine vendors, and patient demographics. This lack of generalisation further reinforces the need for collaborative, privacy-preserving learning approaches. FL not only addresses the legal and ethical limitations of data sharing but also enables the training of models on heterogeneous, multi-institutional datasets [26]. This diversity improves the robustness and adaptability of AI models, particularly when applied to real-world clinical settings that differ from the original training environment [27, 28].

While FL has gained traction in other medical imaging domains—such as MRI, CT, and histopathology—its application in fetal ultrasound remains limited. Most prior studies have focused on common obstetric tasks, such as fetal plane classification or gestational age estimation [29,30]. Very few have addressed the challenge of rare congenital heart defect detection in a federated setting, and to the best of our knowledge, no existing work has applied FL specifically to IAA detection using fetal ultrasound. Given the rarity of the condition and the need for broad data diversity, this application domain represents a critical but underexplored opportunity.

Building upon the aforementioned challenges of data scarcity, privacy constraints, and inter-institutional heterogeneity in rare congenital heart defect screening, the main contributions of this work are summarised as follows:

- (1) We present the first application of federated learning to fetal ultrasound imaging for privacy-preserving prenatal detection of IAA.
- (2) We curate a multi-centre dataset of >4000 heterogeneous fetal echocardiographic images from five clinical sites and conduct a systematic evaluation of three representative federated strategies—FedAvg [31], FedProx [32], and FedBABU [33].
- (3) We comprehensively compare three CNN backbones—ResNet-50 [34], EfficientNet-B3 [35], and DenseNet-121 [36,37]—under federated settings, demonstrating that DenseNet-121 achieves superior cross-site generalisation and recall.
- (4) Our experiments show that the federated framework improves recall and F1-score in data-scarce centres, validating its practical clinical value for rare congenital heart defect screening.

2. Related work

Over the past decade, AI, especially deep learning, has revolutionised prenatal screening, with fetal ultrasound emerging as a key modality for AI-based Computer-Aided Diagnosis (CAD). Recent advances not only focus on standard plane recognition and CHD detection, but also address clinical challenges such as data privacy and inter-institutional variability.

Early research in AI-assisted fetal ultrasound focused on standard plane detection and anatomical landmark localisation. These tasks are fundamental for downstream CHD screening, as accurate identification of standard views such as the four-chamber view (4CV), left ventricular outflow tract (LVOT), and three-vessel trachea (3VT) view provides the structural basis for cardiac assessment. For instance, Baumgartner et al. developed in 2017 a CNN-based pipeline to classify fetal cardiac planes from 2D ultrasound images with high accuracy [38]. Similarly, in 2020, Arnaout et al. proposed a deep learning model trained on thousands of labelled ultrasound images to identify standard views and highlight suspicious anatomical regions [39]. These foundational works demonstrate the potential of deep learning in replicating expert-level anatomical recognition, thus laying the groundwork for automated CHD screening.

While earlier studies have established deep learning models for standard plane classification and common CHD types, the detection of rare defects like IAA remains constrained by limited annotated datasets. In the specific context of CHD detection, deep learning models have been trained to recognise patterns associated with various heart defects. In 2021, Komatsu et al. proposed the SONO framework, which enables robust detection of cardiac structural abnormalities in fetal ultrasound

videos by training only on normal data and employing explainable object detection techniques [40]. More recently, in 2023, Pachiyannan et al. developed the Cardiac Deep Learning Model (CDLM), which utilises maternal clinical and fetal health data to accurately predict the risk of congenital heart disease [41]. However, most of these models are trained on relatively common CHD types and struggle with rarer conditions like IAA, largely due to the scarcity of annotated data. Additionally, the inherent variability in fetal pose, gestational age, and scanning quality introduces noise and heterogeneity into the image data, complicating feature extraction and generalisation.

Addressing the limitations posed by data scarcity and variability remains a significant challenge in medical imaging, particularly for rare diseases. One promising approach involves training models on multi-institutional datasets to capture a broader spectrum of patient demographics and imaging conditions. However, legal and ethical constraints—such as the Health Insurance Portability and Accountability Act (HIPAA) in the United States and the General Data Protection Regulation (GDPR) in the European Union [10]—often restrict the sharing of raw medical images across institutional boundaries. These limitations have historically impeded the scalability of centralised deep learning systems in healthcare.

In response to these challenges, FL has emerged as a viable solution, enabling collaborative model training without necessitating the exchange of patient data. First introduced by Google (McMahan et al. 2017), FL is a privacy-preserving technique for training neural networks on decentralised datasets [31]. In a typical FL framework, each participating institution (client) trains a local model on its private data and transmits only model parameters (e.g., gradients or weights) to a central server. The server then performs a global aggregation of the updates, and this iterative process continues until convergence. This paradigm allows the global model to benefit from diverse data sources while upholding stringent privacy standards.

Since its inception, FL has been successfully applied to a variety of medical imaging tasks. In 2024, Feng proposed a robust FL model based on multicentre CT images to predict high-risk postoperative gastric cancer patients, achieving Area Under the ROC Curve (AUC) up to 0.869 while effectively addressing data privacy and heterogeneity [42]. Hossain designed an FL framework with an Inception-V3 backbone to classify histopathological images of lung and colon cancers, reaching up to 100% accuracy while preserving patient data privacy [43]. Shiri developed a collaborative FL system combining multiple deep learning models (including Vgg19 and DenseNet169) to differentiate COVID-19 from other chest diseases using chest X-rays, overcoming data imbalance and Non-IID challenges and improving diagnostic performance [44]. Joynab introduced CNN-based FL architectures for pap-smear cervical cell image classification, achieving test accuracies of 94.4% under IID and 78.4% under Non-IID settings, enabling multi-institutional training with privacy preservation [45]. Zhou constructed FL models on retrospective data from 12 hospitals to predict 30-day mortality risk in pulmonary thromboembolism patients, with AUC surpassing traditional risk scores and centralised models across various data distributions [46]. Patil applied FL with an Exponential African Pelican Optimisation-based Deep Convolutional Neural Network (DCNN) to detect adolescent depression using speech and MRI data, achieving 98% accuracy and demonstrating the feasibility of multimodal mental health diagnosis [47]. These studies underscore FL's potential to accommodate heterogeneous medical datasets while remaining compliant with data protection regulations.

Despite this progress, the application of FL to fetal ultrasound imaging remains underexplored, particularly in the context of CHD detection. Ultrasound is inherently more variable than modalities like CT or MRI, owing to its susceptibility to artifacts, operator dependency, and fetal movement. This leads to significant inter-institutional variability and poses challenges for robust federated training, as data are often Non-IID.

To address such heterogeneity, several FL optimisation algorithms have been proposed. FedAvg (McMahan et al. 2017), the canonical approach, performs a weighted average of locally trained models. However, it is prone to client drift under Non-IID data distributions. FedProx (Li et al. 2020) extends FedAvg by introducing a proximal term to mitigate divergence in local updates. More recently, FedBABU (Oh et al. 2022) separates the global feature extractor from the local classifier, allowing personalised model adaptation at each site. Despite their promise, these algorithms have not been systematically evaluated in the context of fetal ultrasound or CHD detection, representing a critical gap in current research.

In summary, while deep learning holds considerable promise for CHD detection in fetal ultrasound, its utility for rare conditions such as IAA is constrained by data scarcity and privacy regulations. FL offers a compelling framework for cross-institutional collaboration, but its efficacy under the unique conditions of ultrasound imaging—marked by high variability and Non-IID distributions—requires further investigation. Moreover, the choice of the FL optimisation strategy and backbone architecture, particularly the integration of transfer learning, may significantly influence model performance. This study seeks to address these gaps by designing and evaluating an FL framework tailored to prenatal IAA detection, with systematic comparisons of FL algorithms and transfer learning backbones.

3. Methods

3.1. Proposed model

In this study, we adopt an FL framework to collaboratively train deep learning models on fetal echocardiographic images, aiming to identify the presence of IAA without sharing raw data among multiple institutions. To enhance model performance and generalisation, particularly under data-scarce conditions typical of medical imaging tasks, we incorporate transfer learning by leveraging three widely adopted Convolutional Neural Networks (CNNs) pretrained on the ImageNet dataset—ResNet-50, DenseNet-121, and EfficientNet-B3.

Transfer learning enables the reuse of low- and mid-level visual features learnt from large-scale natural image datasets, significantly accelerating convergence and improving accuracy in downstream tasks with limited labelled data. In our approach, fetal ultrasound images are first preprocessed through a series of operations including resizing, normalisation, and augmentation (as detailed in Section 3.2), and then individually fed into one of the selected backbone networks. These models act as local clients within the federated setup, where training occurs locally and only model weights are periodically aggregated on the central server. Each backbone network independently extracts high-dimensional feature maps, which are subsequently processed by a shared classification head for binary classification (IAA present vs. absent).

The architecture of each backbone model contains distinct structural modules that contribute to their respective strengths in medical image analysis. As shown in Fig. 1, we highlight three representative modules: the Mobile Inverted Bottleneck Convolution (MBConv) block from EfficientNet-B3 (Fig. 1 (a)), the Dense Block from DenseNet-121 (Fig. 1 (b)), and the Residual Block from ResNet-50 (Fig. 1 (c)).

The MBConv block, integral to EfficientNet-B3, is designed for computational efficiency and feature expressiveness. It consists of a 1×1 pointwise convolution for channel reduction, followed by a depth-wise separable convolution, and then a Squeeze-and-Excitation (SE) module to adaptively recalibrate channel-wise feature responses. The block concludes with another 1×1 convolution for channel expansion. Swish activation is employed to enhance gradient flow and non-linearity, while residual connections ensure stable backpropagation. This design enables EfficientNet-B3 to maintain high accuracy with significantly fewer parameters, making it well-suited for federated deployment scenarios.

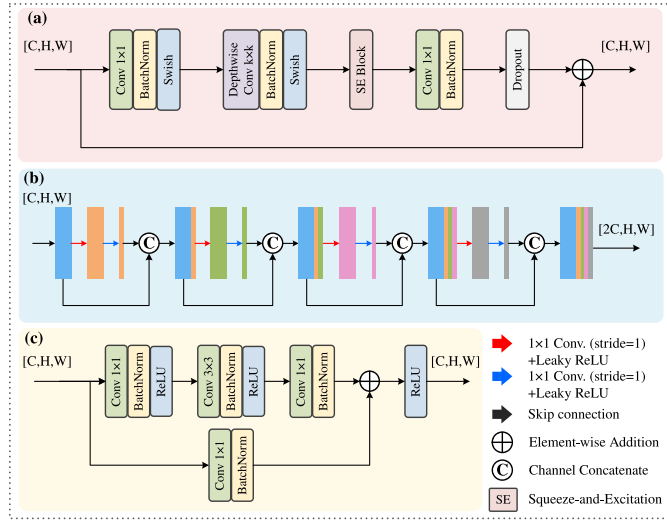


Fig. 1. The internal structures of three representative CNN modules used as building blocks in the backbone networks: (a) MBConv block, (b) Dense block, (c) Residual block.

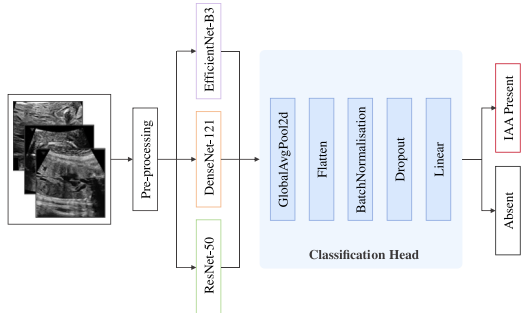


Fig. 2. The overall pipeline of the proposed FL framework for detecting IAA in fetal ultrasound images.

DenseNet-121 introduces a radically different connectivity pattern through its Dense Blocks, where each layer receives input from all preceding layers via channel-wise concatenation. This dense connectivity promotes feature reuse, strengthens gradient propagation, and reduces the number of required parameters, which is particularly beneficial in medical imaging tasks involving small datasets. Each layer typically consists of Batch Normalisation, Rectified Linear Unit (ReLU) activation, and 1×1 followed by 3×3 convolutional layers. The architectural emphasis on information flow and feature accumulation helps DenseNet perform robustly in complex classification problems.

ResNet-50, on the other hand, employs Residual Blocks that leverage shortcut connections to enable the direct propagation of input features across layers. This approach effectively mitigates the vanishing gradient problem and facilitates the training of very deep networks. Each block comprises a bottleneck structure with 1×1 , 3×3 , and another 1×1 convolution, interspersed with Batch Normalisation and ReLU activations. A skip connection adds the block's input to its output before the final activation, preserving both low-level and high-level features and enhancing representational capacity.

All three backbones are followed by a unified classification head, as depicted in Fig. 2. This component begins with a Global Average Pooling (GlobalAvgPool2d) layer to reduce spatial dimensions, followed by a Flatten operation to convert the output into a 1D vector. Subsequent Batch Normalisation improves training stability, and Dropout serves to prevent overfitting. Finally, a fully connected layer produces a binary output indicating the presence or absence of IAA. This head

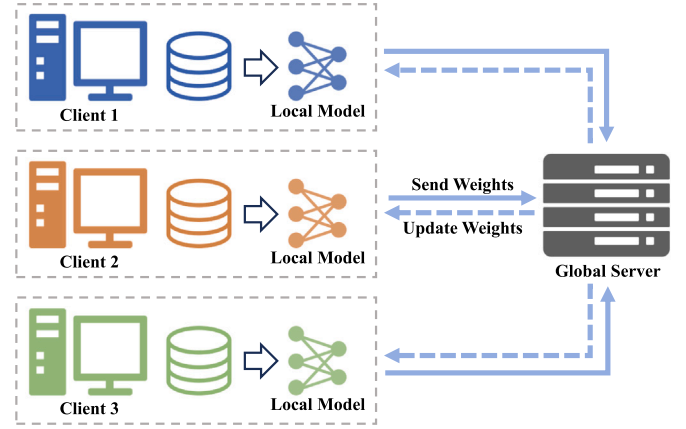


Fig. 3. Structure of the FL framework.

design ensures consistency across different backbones and facilitates fair performance comparison in the FL context.

By integrating pretrained CNN architectures with distinct structural advantages into a unified and privacy-preserving FL framework, our approach effectively addresses several key challenges in medical image analysis. These include limited data availability, the need for efficient training under computational constraints, and the strict requirement for data confidentiality across institutions. The modular design of each backbone — incorporating residual, dense, or inverted bottleneck connections — provides diverse feature representation capabilities, which are crucial for capturing the subtle anatomical patterns in fetal echocardiographic images. Moreover, the shared classification head ensures architectural consistency across different models, enabling the framework to remain extensible and adaptable to future model variants or other cardiac anomaly detection tasks. This model design lays a strong foundation for scalable, collaborative deep learning in sensitive medical imaging applications.

3.2. Federated learning architecture

The workflow of FL is shown in Fig. 3. We compared three FL methods in this study to determine which is more suitable for this multi-centre study.

- **FedAvg** [31]: FedAvg is a classical FL algorithm where the server aggregates model parameters via averaging.
- **FedProx** [32]: FedProx enhances FedAvg by introducing proximal term constraints to regularise client models parameters' deviation from global model parameters, alleviating performance issues under Non-IID settings.
- **FedBABU** [33]: FedBABU decouples model parameters into globally shared layers and client-specific task layers, aggregating only the former to address Non-IID challenges.

4. Experiments and results

4.1. Datasets

We retrospectively collected 2D fetal echocardiographic ultrasound images from three medical centres between March 2014 and December 2024, covering fetuses with gestational ages ranging from 19.3 to 36.6 weeks. The dataset includes two standard views: the three-vessel and trachea (3VT) view and the aortic arch (AA) view, as recommended by the Guidelines and Recommendations for Performance of the Fetal Echocardiogram (American Society of Echocardiography, 2020 update). All images were de-identified and collected under institutional ethics approvals. The participating centres, case distributions,

Table 1
Details of the multicentre dataset.

Medical centre	Location	Normal cases	IAA cases	Ultrasound machines
Beijing Anzhen Hospital, Capital Medical University	Beijing, China	3378	474	GE Voluson E8 (GE Healthcare, Chicago, USA); GE Voluson E10 (GE Healthcare, Wuxi, China)
Peking University First Hospital – Ningxia Women and Children's Hospital	Yinchuan, Ningxia, China	68	64	GE Voluson E10 (GE Healthcare, Wuxi, China)
Fujian Maternity and Child Health Hospital, Fujian Medical University	Fuzhou, Fujian, China	80	80	GE Voluson E8 (Chicago, USA); GE Voluson E10 (Wuxi, China); HERA W10 (Samsung, Germany); PHILIPS (Philips Healthcare, Amsterdam, Netherlands)

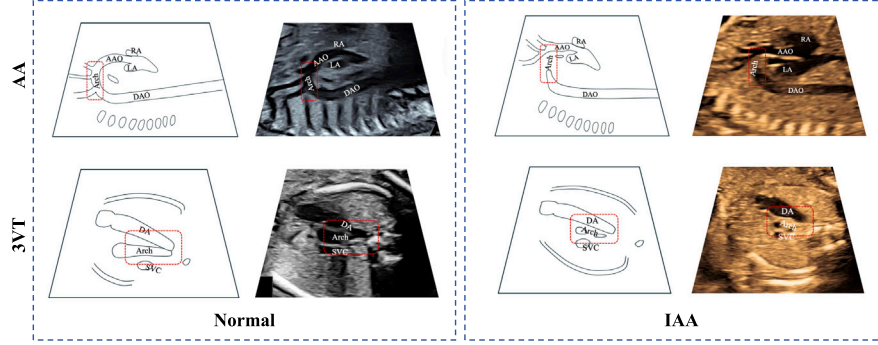


Fig. 4. Comparative ultrasound features in our dataset: Normal vs. IAA. Left: Schematic diagrams. Right: 2D ultrasound images (red boxes: lesions). Top: AA view showing aortic arch continuity (Normal) vs. discontinuity (IAA). Bottom: 3VT view demonstrating vessel alignment. LA: Left atrium. RA: Right atrium. AAO: Ascending aorta. DAO: Descending aorta. Arch: Aortic arch. DA: Ductus arteriosus. SVC: Superior vena cava.

and ultrasound machines used are as shown in Table 1. Differences in case distribution among centres were assessed using Pearson's chi-square test, which revealed a significant association between centre and disease status (Pearson's $\chi^2 = 293.12$, $df = 2$, $p < 0.001$). The Fig. 4 illustrates the anatomical regions where the lesions are located.

To address sample imbalance due to the large number of images from the Beijing Anzhen Hospital, its dataset was divided into three independent subsets. Each was treated as a separate centre during training and evaluation, resulting in a total of five centres included in the study.

All images were initially stored in DICOM format and were subsequently converted to PNG format for model development. Each image was reviewed and annotated by experienced fetal cardiologists or senior sonographers to ensure diagnostic accuracy. This work involved human subjects in its research. Approval of all ethical and experimental procedures and protocols was granted by the Ethics Committee for Biomedical Research at the Beijing Anzhen Hospital under Application KS2025142.

4.2. Data preprocessing

To prevent overfitting, increase data diversity, and enhance the generalisation ability of the model, a series of preprocessing steps were applied to the fetal echocardiographic images prior to training. These steps include image filtering, size normalisation, data augmentation, and dataset partitioning, as outlined below:

- Image filtering and region of interest (ROI) extraction: Low-quality images with severe blurring, non-standard views, or occlusions were removed. The ROI, mainly focusing on the aortic arch and related cardiac structures, was extracted through central cropping or semi-automatic methods.
- Size normalisation and grayscale normalisation: All images were resized to 224×224 pixels, converted to grayscale, and normalised to standardise pixel intensity distribution and improve training stability.

- Data augmentation: To expand the training dataset and enhance the model's robustness to anatomical and acquisition variability, we employed data augmentation techniques including random rotation within ± 30 degrees and brightness adjustment by a factor randomly sampled from the range $[0.8, 1.2]$. Horizontal flipping was excluded due to the asymmetric anatomical structure of cardiac images.
- Label encoding and dataset splitting: Images were labelled as 0 (normal) or 1 (IAA).

4.3. Evaluation metrics

In this study, we selected four evaluation metrics — Precision, Recall, F1-score, and Specificity — to assess the performance of our binary classification model for detecting IAA. Recall measures the model's ability to identify true IAA cases, while specificity reflects its accuracy in recognising normal cases, together ensuring a balanced consideration of both missed and false diagnoses. Precision indicates the proportion of correctly identified abnormal cases among all cases predicted as abnormal, and the F1-score provides a harmonic mean of precision and recall [48], which is particularly relevant for class-imbalanced medical imaging tasks. Taken together, these metrics offer a comprehensive evaluation of the effectiveness and reliability of the model in clinical screening, highlighting its value in both disease detection and the reduction of false positives. The equations of these metrics are as follows:

$$Precision = \frac{TP}{TP + FP} \quad (1)$$

$$Recall = \frac{TP}{TP + FN} \quad (2)$$

$$F1 - score = \frac{2 * Precision * Recall}{Precision + Recall} \quad (3)$$

$$Specificity = \frac{TN}{TN + FP} \quad (4)$$

4.4. Experimental setup

This study systematically evaluated the performance of three mainstream convolutional neural networks — ResNet-50, EfficientNet-B3, and DenseNet-121 — on the screening of IAA using ultrasound image data from five clinical centres (Anzhen 1, Anzhen 2, Anzhen 3, Fujian, and Ningxia). We assessed each model under local independent learning (Local Learning) and compared three FL strategies (FedAvg, FedBABU, and FedProx) in a multi-centre collaborative setting. To prevent variations in data distribution arising from dataset splitting from interfering with model performance, we employed five-fold cross-validation for model training, with federated learning conducted on each fold. The experiments focused on Precision, Recall, F1-score, and Specificity [49–51], aiming to comprehensively reflect the models' ability to detect IAA-positive cases and maintain classification balance. Results are shown in Tables 2 and 3.

4.5. Result analysis

4.5.1. Performance comparison: Federated vs. Local learning

As presented in Tables 2 and 3, Federated Learning (FL) strategies demonstrated comprehensive superiority over Local Learning across most centres and metrics. This advantage is particularly evident in metrics sensitive to false negatives, such as Recall and F1-score. For instance, in the Anzhen 1 centre, the Recall of ResNet-50 improved from 0.480 ± 0.172 under local training to 0.753 ± 0.063 with FedAvg. Similarly, at the Ningxia centre, the DenseNet-121 model saw its Recall surge from 0.629 ± 0.144 (Local) to 0.964 ± 0.027 (FedAvg), with the F1-score rising from 0.753 to 0.926.

Crucially, beyond performance gains, FL significantly enhanced model stability. The standard deviations of the results in Table 3 are consistently lower than those in Table 2. For example, the standard deviation for ResNet-50's Recall at Fujian decreased from 0.190 (Local) to 0.065 (FedAvg). This indicates that collaborative training effectively mitigates the bias and variance caused by limited sample sizes and data heterogeneity in individual centres, leading to more robust generalisation for rare disease detection like IAA.

4.5.2. Comparison of federated learning strategies

Among the three FL frameworks, FedAvg and FedProx exhibited comparable top-tier performance, yet a granular comparison reveals subtle differences. Although FedProx is theoretically designed to handle data heterogeneity via a proximal term, FedAvg generally achieved slightly higher F1-scores in this study, particularly in centres with better data quality. For example, with the DenseNet-121 model at Anzhen 1, FedAvg reached an F1-score of 0.865, slightly surpassing FedProx's 0.856. This suggests that the strict constraint of the proximal term in FedProx might have slightly limited the model's convergence peak in relatively homogeneous data partitions. However, FedProx demonstrated its robustness in specific challenging scenarios; for instance, in the Fujian centre with the EfficientNet-B3 model, FedProx achieved a higher F1-score (0.629) compared to FedAvg (0.598). Referring to Table 4, although FedProx introduces an extra calculation for the proximal term, the computational overhead is negligible. Therefore, while FedAvg is empirically slightly superior and more efficient for this specific task, FedProx remains a robust alternative when facing severe data heterogeneity.

In contrast, FedBABU (updates only the body) underperformed compared to the end-to-end update strategies. As shown in Table 3, FedBABU suffered from a notable drop in Precision and Specificity, leading to lower F1-scores (e.g., Anzhen 1 ResNet-50 F1: 0.753 vs. FedAvg 0.842). This suggests that for complex medical imaging tasks with domain shifts, fine-tuning the classifier head alone may be insufficient, and updating the backbone feature extractor is crucial for capturing centre-specific variations.

4.5.3. Backbone architecture and efficiency analysis

Comparing the backbone networks, DenseNet-121 emerged as the most optimal choice, balancing performance and efficiency.

(1) Performance: DenseNet-121 consistently outperformed ResNet-50 and EfficientNet-B3 in F1-score and Recall across both local and federated settings. EfficientNet-B3, while computationally light, struggled with Recall (e.g., 0.421 in Anzhen 1 under FedAvg), potentially due to model capacity constraints in capturing subtle IAA features.

(2) Efficiency: Referring to Table 4, DenseNet-121 has the fewest parameters (~6.95M) compared to ResNet-50 (~23.5M) and EfficientNet (~10.7M). This makes DenseNet-121 highly communication-efficient for FL, requiring significantly less bandwidth per global round while delivering the highest diagnostic accuracy.

4.5.4. Interpretability and visualisation

To validate the clinical relevance of our method, we visualised the model's focus using Gradient-weighted Class Activation Mapping (Grad-CAM). As shown in Fig. 6, the heatmaps generated by the FedAvg-trained DenseNet-121 align closely with the Regions of Interest (ROI) marked by radiologists (Fig. 4). This confirms that the model bases its predictions on pathological features rather than background artifacts, ensuring the reliability required for medical AI systems. Normalised confusion matrices (Fig. 5) further corroborate the model's balanced classification capability.

5. Discussion

5.1. Performance and reliability of federated learning strategies

The experimental results demonstrate that FL offers a substantial improvement over local learning, particularly in enhancing the model's sensitivity to positive cases (Recall) and overall balance (F1-score). This advantage is most pronounced in centres with limited data volume or skewed distributions, such as Anzhen 1 and Ningxia. Crucially, beyond the improvements in mean performance metrics, FL significantly reduces the standard deviation of the results across repeated experiments. This reduction in variance indicates that collaborative training effectively mitigates the instability often associated with training deep learning models on small, isolated medical datasets. By aggregating knowledge from multiple institutions, FL provides a regularisation effect that prevents overfitting to local noise, thereby delivering not only more accurate but also more clinically reliable diagnostic tools.

5.2. Comparative analysis of federated learning frameworks

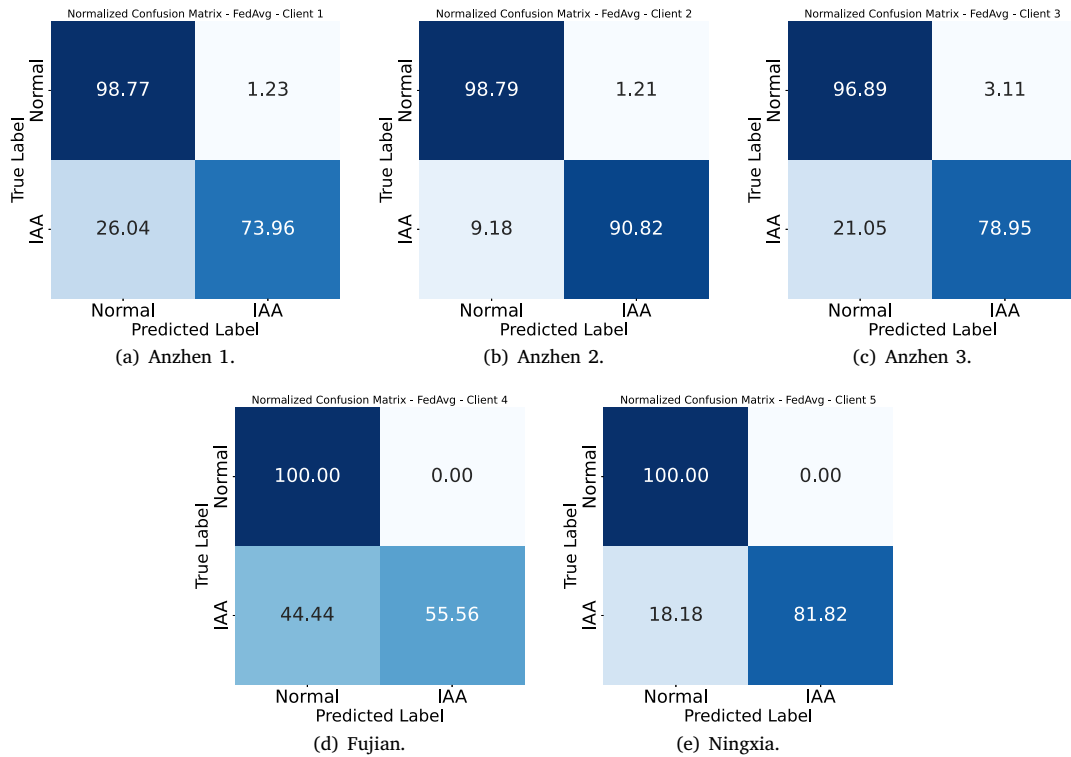
Among the evaluated frameworks, FedAvg and FedProx consistently outperformed FedBABU. While FedProx is theoretically designed to address data heterogeneity via a proximal term, our empirical results show that FedAvg achieved slightly superior F1-scores in most scenarios (e.g., DenseNet-121 at Anzhen 1). This suggests that for the degree of statistical heterogeneity present in our dataset, the standard averaging mechanism of FedAvg is sufficiently robust. The proximal constraint in FedProx might have imposed a slight restriction on the model's convergence towards the optimal solution in this specific task. However, given that Table 4 indicates the extra computational overhead of FedProx is negligible, it remains a viable alternative for scenarios with more extreme Non-IID data.

In contrast, the inferior performance of FedBABU highlights a critical insight: for medical imaging tasks involving domain shifts (e.g., different scanner protocols across centres), updating the backbone (feature extractor) is essential. FedBABU's strategy of freezing the backbone during local updates likely prevented the model from learning centre-invariant representations effectively, leading to the observed drop in Specificity and Precision.

Table 2

Experiments concerning local learning. The epochs count of local learning is 50; the learning rate of each models is 0.0001; the batch size is 64. The values behind $\pm$ is the standard deviation of the results.

Centre	Model	Precision	Recall	F1-score	Specificity
Anzhen 1	ResNet-50	0.977 $\pm$ 0.015	0.480 $\pm$ 0.172	0.626 $\pm$ 0.141	0.992 $\pm$ 0.012
	EfficientNet-B3	0.623 $\pm$ 0.043	0.445 $\pm$ 0.099	0.509 $\pm$ 0.063	0.818 $\pm$ 0.064
	DenseNet-121	0.988 $\pm$ 0.013	0.606 $\pm$ 0.089	0.747 $\pm$ 0.063	0.995 $\pm$ 0.007
Anzhen 2	ResNet-50	0.984 $\pm$ 0.007	0.539 $\pm$ 0.123	0.688 $\pm$ 0.108	0.994 $\pm$ 0.004
	EfficientNet-B3	0.729 $\pm$ 0.066	0.519 $\pm$ 0.083	0.601 $\pm$ 0.056	0.860 $\pm$ 0.046
	DenseNet-121	0.991 $\pm$ 0.018	0.616 $\pm$ 0.048	0.758 $\pm$ 0.034	0.996 $\pm$ 0.009
Anzhen 3	ResNet-50	0.992 $\pm$ 0.010	0.567 $\pm$ 0.137	0.711 $\pm$ 0.108	0.997 $\pm$ 0.004
	EfficientNet-B3	0.654 $\pm$ 0.081	0.523 $\pm$ 0.061	0.574 $\pm$ 0.031	0.832 $\pm$ 0.068
	DenseNet-121	0.989 $\pm$ 0.006	0.661 $\pm$ 0.079	0.789 $\pm$ 0.057	0.996 $\pm$ 0.009
Fujian	ResNet-50	0.873 $\pm$ 0.033	0.522 $\pm$ 0.190	0.633 $\pm$ 0.148	0.926 $\pm$ 0.035
	EfficientNet-B3	0.502 $\pm$ 0.094	0.306 $\pm$ 0.095	0.374 $\pm$ 0.092	0.709 $\pm$ 0.100
	DenseNet-121	0.950 $\pm$ 0.047	0.544 $\pm$ 0.254	0.641 $\pm$ 0.271	0.963 $\pm$ 0.036
Ningxia	ResNet-50	0.988 $\pm$ 0.015	0.657 $\pm$ 0.152	0.779 $\pm$ 0.108	0.991 $\pm$ 0.011
	EfficientNet-B3	0.714 $\pm$ 0.230	0.461 $\pm$ 0.209	0.549 $\pm$ 0.208	0.839 $\pm$ 0.098
	DenseNet-121	0.977 $\pm$ 0.028	0.629 $\pm$ 0.144	0.753 $\pm$ 0.118	0.982 $\pm$ 0.022

**Fig. 5.** Normalised confusion matrices of each client of FedAvg.

5.3. Limitations and future directions

In this study, the available data consisted solely of two classes: healthy and IAA. As a result, the trained models are limited to providing an assessment of the presence or absence of IAA. From a clinical perspective, however, simply determining whether a fetus has IAA is insufficient. Future work should focus on refining the dataset to distinguish between specific subtypes of IAA. Furthermore, as the number of categories increases, the distributional differences among data from different centres may also become more pronounced. To address this, it will be necessary to optimise the FL framework to better handle challenges such as Non-IID data [18]. Finally, although we analysed communication volume, future deployments should also consider differential privacy mechanisms to further rigorously quantify and protect patient privacy boundaries.

6. Conclusion

In this study, we presented the first application of FL to prenatal detection of IAA using fetal ultrasound imaging. Through a comprehensive evaluation of three FL strategies and three CNN architectures, we demonstrated that FL enables substantially improving diagnostic performance—particularly in settings with limited data availability. Beyond merely improving mean accuracy, our results show that FL significantly enhances clinical reliability by reducing the variance of model performance across repeated experiments.

Our comparative analysis offers two critical guidelines for deploying FL in medical imaging. First, regarding model selection, DenseNet-121 emerged as the optimal architecture, achieving the best trade-off between diagnostic sensitivity (Recall) and communication efficiency (lowest parameter count), making it highly suitable for bandwidth-constrained hospital networks. Second, regarding FL strategies, the

Table 3

Experiments of federated learning. The global round count of each FL framework is 50; epochs per round of local update are 1; the learning rate of each model is 0.0001; the batch size is 64. The values behind $\pm$ is the standard deviation of the results.

Centre	Model	Precision	Recall	F1-score	Specificity
FedAvg					
Anzhen 1	ResNet-50	0.960 $\pm$ 0.010	0.753 $\pm$ 0.063	0.842 $\pm$ 0.039	0.979 $\pm$ 0.006
	EfficientNet-B3	0.967 $\pm$ 0.022	0.421 $\pm$ 0.045	0.584 $\pm$ 0.041	0.990 $\pm$ 0.007
	DenseNet-121	0.951 $\pm$ 0.021	0.794 $\pm$ 0.030	0.865 $\pm$ 0.015	0.973 $\pm$ 0.013
Anzhen 2	ResNet-50	0.969 $\pm$ 0.012	0.758 $\pm$ 0.037	0.850 $\pm$ 0.027	0.983 $\pm$ 0.007
	EfficientNet-B3	0.951 $\pm$ 0.020	0.430 $\pm$ 0.023	0.591 $\pm$ 0.020	0.984 $\pm$ 0.007
	DenseNet-121	0.960 $\pm$ 0.008	0.797 $\pm$ 0.030	0.871 $\pm$ 0.018	0.977 $\pm$ 0.004
Anzhen 3	ResNet-50	0.959 $\pm$ 0.017	0.771 $\pm$ 0.045	0.854 $\pm$ 0.028	0.981 $\pm$ 0.008
	EfficientNet-B3	0.947 $\pm$ 0.023	0.491 $\pm$ 0.032	0.646 $\pm$ 0.029	0.984 $\pm$ 0.007
	DenseNet-121	0.948 $\pm$ 0.022	0.824 $\pm$ 0.030	0.881 $\pm$ 0.014	0.974 $\pm$ 0.012
Fujian	ResNet-50	0.795 $\pm$ 0.043	0.794 $\pm$ 0.065	0.794 $\pm$ 0.046	0.804 $\pm$ 0.052
	EfficientNet-B3	0.841 $\pm$ 0.083	0.472 $\pm$ 0.091	0.598 $\pm$ 0.082	0.910 $\pm$ 0.055
	DenseNet-121	0.794 $\pm$ 0.028	0.817 $\pm$ 0.076	0.804 $\pm$ 0.047	0.799 $\pm$ 0.027
Ningxia	ResNet-50	0.896 $\pm$ 0.038	0.915 $\pm$ 0.026	0.905 $\pm$ 0.027	0.893 $\pm$ 0.043
	EfficientNet-B3	0.785 $\pm$ 0.072	0.598 $\pm$ 0.089	0.678 $\pm$ 0.083	0.839 $\pm$ 0.043
	DenseNet-121	0.891 $\pm$ 0.045	0.964 $\pm$ 0.027	0.926 $\pm$ 0.028	0.880 $\pm$ 0.053
FedBABU					
Anzhen 1	ResNet-50	0.815 $\pm$ 0.109	0.718 $\pm$ 0.095	0.753 $\pm$ 0.062	0.878 $\pm$ 0.094
	EfficientNet-B3	0.413 $\pm$ 0.051	0.567 $\pm$ 0.112	0.532 $\pm$ 0.029	0.566 $\pm$ 0.083
	DenseNet-121	0.790 $\pm$ 0.072	0.758 $\pm$ 0.147	0.758 $\pm$ 0.147	0.855 $\pm$ 0.068
Anzhen 2	ResNet-50	0.673 $\pm$ 0.189	0.835 $\pm$ 0.240	0.686 $\pm$ 0.112	0.590 $\pm$ 0.307
	EfficientNet-B3	0.490 $\pm$ 0.031	0.587 $\pm$ 0.070	0.532 $\pm$ 0.029	0.566 $\pm$ 0.083
	DenseNet-121	0.674 $\pm$ 0.151	0.778 $\pm$ 0.335	0.627 $\pm$ 0.215	0.635 $\pm$ 0.227
Anzhen 3	ResNet-50	0.810 $\pm$ 0.179	0.710 $\pm$ 0.236	0.698 $\pm$ 0.136	0.841 $\pm$ 0.185
	EfficientNet-B3	0.429 $\pm$ 0.092	0.571 $\pm$ 0.139	0.463 $\pm$ 0.046	0.498 $\pm$ 0.225
	DenseNet-121	0.753 $\pm$ 0.225	0.653 $\pm$ 0.278	0.607 $\pm$ 0.127	0.731 $\pm$ 0.368
Fujian	ResNet-50	0.618 $\pm$ 0.103	0.744 $\pm$ 0.331	0.602 $\pm$ 0.186	0.465 $\pm$ 0.349
	EfficientNet-B3	0.561 $\pm$ 0.023	0.672 $\pm$ 0.201	0.599 $\pm$ 0.093	0.503 $\pm$ 0.137
	DenseNet-121	0.660 $\pm$ 0.152	0.778 $\pm$ 0.182	0.680 $\pm$ 0.040	0.515 $\pm$ 0.318
Ningxia	ResNet-50	0.849 $\pm$ 0.135	0.559 $\pm$ 0.345	0.568 $\pm$ 0.288	0.827 $\pm$ 0.162
	EfficientNet-B3	0.472 $\pm$ 0.129	0.474 $\pm$ 0.204	0.463 $\pm$ 0.160	0.517 $\pm$ 0.152
	DenseNet-121	0.775 $\pm$ 0.233	0.609 $\pm$ 0.374	0.541 $\pm$ 0.239	0.600 $\pm$ 0.448
FedProx					
Anzhen 1	ResNet-50	0.977 $\pm$ 0.010	0.700 $\pm$ 0.066	0.814 $\pm$ 0.046	0.989 $\pm$ 0.005
	EfficientNet-B3	0.959 $\pm$ 0.025	0.405 $\pm$ 0.071	0.565 $\pm$ 0.064	0.987 $\pm$ 0.009
	DenseNet-121	0.962 $\pm$ 0.011	0.771 $\pm$ 0.032	0.856 $\pm$ 0.019	0.980 $\pm$ 0.006
Anzhen 2	ResNet-50	0.971 $\pm$ 0.009	0.713 $\pm$ 0.033	0.822 $\pm$ 0.024	0.985 $\pm$ 0.005
	EfficientNet-B3	0.956 $\pm$ 0.024	0.423 $\pm$ 0.039	0.585 $\pm$ 0.035	0.986 $\pm$ 0.009
	DenseNet-121	0.967 $\pm$ 0.010	0.775 $\pm$ 0.017	0.860 $\pm$ 0.014	0.981 $\pm$ 0.005
Anzhen 3	ResNet-50	0.982 $\pm$ 0.011	0.720 $\pm$ 0.042	0.830 $\pm$ 0.025	0.992 $\pm$ 0.005
	EfficientNet-B3	0.954 $\pm$ 0.010	0.491 $\pm$ 0.049	0.646 $\pm$ 0.041	0.986 $\pm$ 0.004
	DenseNet-121	0.954 $\pm$ 0.016	0.810 $\pm$ 0.029	0.876 $\pm$ 0.016	0.976 $\pm$ 0.008
Fujian	ResNet-50	0.741 $\pm$ 0.062	0.806 $\pm$ 0.063	0.772 $\pm$ 0.058	0.729 $\pm$ 0.075
	EfficientNet-B3	0.855 $\pm$ 0.056	0.506 $\pm$ 0.097	0.629 $\pm$ 0.078	0.915 $\pm$ 0.045
	DenseNet-121	0.725 $\pm$ 0.053	0.844 $\pm$ 0.045	0.780 $\pm$ 0.047	0.693 $\pm$ 0.068
Ningxia	ResNet-50	0.890 $\pm$ 0.023	0.933 $\pm$ 0.037	0.910 $\pm$ 0.022	0.884 $\pm$ 0.026
	EfficientNet-B3	0.853 $\pm$ 0.073	0.687 $\pm$ 0.107	0.759 $\pm$ 0.087	0.884 $\pm$ 0.051
	DenseNet-121	0.881 $\pm$ 0.041	0.973 $\pm$ 0.017	0.924 $\pm$ 0.027	0.866 $\pm$ 0.050

Table 4

Approximate computational and communication characteristics of different federated learning strategies and backbone networks.

FL strategy	Backbone	Approx. model parameters	Parameters communicated per global round	Extra computational overhead
FedAvg	ResNet-50	~23.5M	All model parameters	No
FedAvg	EfficientNet	~10.7M	All model parameters	No
FedAvg	DenseNet-121	~6.95M	All model parameters	No
FedProx	ResNet-50	~23.5M	All model parameters	Negligible (proximal term)
FedProx	EfficientNet	~10.7M	All model parameters	Negligible
FedProx	DenseNet-121	~6.95M	All model parameters	Negligible
FedBABU	ResNet-50	~23.5M	Feature extractor only	No
FedBABU	EfficientNet	~10.7M	Feature extractor only	No
FedBABU	DenseNet-121	~6.95M	Feature extractor only	No

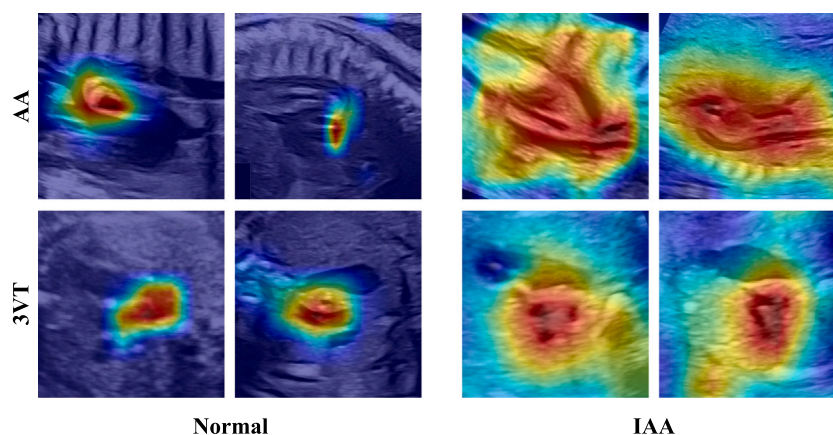


Fig. 6. Gradient-weighted class activation mapping visualisation for DenseNet-based fetal ultrasound classification. Comparative Grad-CAM Heatmaps: DenseNet's Attention Patterns in AA and 3VT ultrasound planes for normal (left) vs. IAA (right) cases (red regions indicate high model attention for classification decisions).

superior performance of FedAvg and FedProx over FedBABU underscores that updating the feature extractor (backbone) is essential to cope with the domain shifts inherent in multi-centre ultrasound data.

Our work not only validated the effectiveness of FL for collaborative training in fetal cardiac imaging but also filled a critical gap in the literature by pioneering its use for IAA detection. This novel application highlights the potential of federated AI to transform rare disease screening by overcoming data scarcity and privacy barriers. In future work, we aim to extend the framework to include IAA subtype classification and further optimise FL algorithms to better handle Non-IID data and imbalanced class distributions in real-world clinical environments.

CRediT authorship contribution statement

Jiancheng Han: Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Data curation, Conceptualization. **Heqing Wang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Yifan Feng:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Conceptualization. **Qi Yang:** Writing – review & editing, Resources, Data curation, Conceptualization. **Jingtan Li:** Writing – review & editing, Investigation. **Haojie Zhang:** Writing – review & editing, Investigation. **Yihua He:** Supervision, Resources, Project administration, Funding acquisition, Data curation. **Jiang Liu:** Writing – review & editing, Supervision. **Toru Nakamura:** Writing – review & editing, Supervision. **Yang Cao:** Writing – review & editing, Supervision. **Naidi Sun:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Kun Qian:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Bin Hu:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Xinru Gao:** Resources, Data curation. **Yan Xia:** Resources, Data curation. **Zongjie Weng:** Resources, Data curation. **Björn W. Schuller:** Writing – review & editing, Supervision. **Yoshiharu Yamamoto:** Writing – review & editing, Supervision.

Declaration of competing interest

There is no conflict of interest among all the authors.

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Data availability

The authors do not have permission to share data.

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